1. Creating and Renaming Files/Directories

1. Create a directory named test_dir using `mkdir`.
2. Inside test_dir, create an empty file called `example.txt`.
3. Rename example.txt to renamed_example.txt using `mv`.

```
rohitbhupat@Rohits-MacBook-Air Documents % cd sahil_devops
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops % mkdir test_dir
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops % cd test_dir
rohitbhupat@Rohits-MacBook-Air test_dir %
rohitbhupat@Rohits-MacBook-Air test_dir % touch example.txt
rohitbhupat@Rohits-MacBook-Air test_dir %
rohitbhupat@Rohits-MacBook-Air test_dir % ls -ltr
total 0
-rw-r--r--  1 rohitbhupat  staff  0 16 Oct 10:41 example.txt
rohitbhupat@Rohits-MacBook-Air test_dir %
rohitbhupat@Rohits-MacBook-Air test_dir %
rohitbhupat@Rohits-MacBook-Air test_dir % mv example.txt renamed_example.txt
rohitbhupat@Rohits-MacBook-Air test_dir %
rohitbhupat@Rohits-MacBook-Air test_dir % ls -ltr
total 0
-rw-r--r--  1 rohitbhupat  staff  0 16 Oct 10:41 renamed_example.txt
```

2. Viewing File Contents

1. Use `cat` to display the contents of /etc/passwd.
2. Display only the first 5 lines of /etc/passwd using `head`.
3. Display only the last 5 lines of /etc/passwd using `tail`.

```
rohitbhupat@Rohits-MacBook-Air test_dir % cat /etc/passwd
##
# User Database
#
# Note that this file is consulted directly only when the system is running
# in single-user mode.  At other times this information is provided by
# Open Directory.
#
# See the opendirectoryd(8) man page for additional information about
# Open Directory.
##
nobody:*:-2:-2:Unprivileged User:/var/empty:/usr/bin/false
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
_uucp:*:4:4:Unix to Unix Copy Protocol:/var/spool/uucp:/usr/sbin/uucico
_taskgated:*:13:13:Task Gate Daemon:/var/empty:/usr/bin/false
_networkd:*:24:24:Network Services:/var/networkd:/usr/bin/false
_installassistant:*:25:25:Install Assistant:/var/empty:/usr/bin/false
_lp:*:26:26:Printing Services:/var/spool/cups:/usr/bin/false
_postfix:*:27:27:Postfix Mail Server:/var/spool/postfix:/usr/bin/false
_scsd:*:31:31:Service Configuration Service:/var/empty:/usr/bin/false
_ces:*:32:32:Certificate Enrollment Service:/var/empty:/usr/bin/false
_appstore:*:33:33:Mac App Store Service:/var/db/appstore:/usr/bin/false
_mcxalr:*:54:54:MCX AppLaunch:/var/empty:/usr/bin/false
_appleevents:*:55:55:AppleEvents Daemon:/var/empty:/usr/bin/false
```

```
rohitbhupat@Rohits-MacBook-Air test_dir % tail -5 /etc/passwd
_spinandd:*:305:305:SPINAND Daemon:/var/db/spinandd:/usr/bin/false
_corespeechd:*:306:306:CoreSpeech Services:/var/empty:/usr/bin/false
_diagnosticservicesd:*:307:307:Diagnostic Services:/var/empty:/usr/bin/false
_mds_stores:*:308:308:Spotlight File Metadata Index Daemon:/var/empty:/usr/bin/false
_oahd:*:441:441:OAH Daemon:/var/empty:/usr/bin/false
rohitbhupat@Rohits-MacBook-Air test_dir %
rohitbhupat@Rohits-MacBook-Air test_dir %
rohitbhupat@Rohits-MacBook-Air test_dir % head -5 /etc/passwd
##
# User Database
#
# Note that this file is consulted directly only when the system is running
# in single-user mode.  At other times this information is provided by
rohitbhupat@Rohits-MacBook-Air test_dir %
```

3. Searching for Patterns

1. Use `grep` to find all lines containing the word “root” in /etc/passwd.

```
rohitbhupat@Rohits-MacBook-Air test_dir % grep "root" /etc/passwd
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
_cvmsroot:*:212:212:CVMS Root:/var/empty:/usr/bin/false
rohitbhupat@Rohits-MacBook-Air test_dir %
```

4. Zipping and Unzipping

1. Compress the test_dir directory into a file named test_dir.zip using `zip`.
2. Unzip test_dir.zip into a new directory named `unzipped_dir`.

```
rohitbhupat@Rohits-MacBook-Air sahil_devops % zip -r test_dir.zip test_dir
updating: test_dir/ (stored 0%)
  adding: test_dir/renamed_example.txt (stored 0%)
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops % ls -ltr
total 8
drwxr-xr-x  2 rohitbhupat  staff   64 16 Oct 10:33 Linux_Basics_Sahil_Shaikh
drwxr-xr-x  3 rohitbhupat  staff   96 16 Oct 10:41 test_dir
-rw-r--r--  1 rohitbhupat  staff  352 16 Oct 10:47 test_dir.zip
rohitbhupat@Rohits-MacBook-Air sahil_devops %
```

5. Downloading Files

1. Use `wget` to download a file from a URL (e.g., https://example.com/sample.txt).

```
rohitbhupat@Rohits-MacBook-Air sahil_devops % wget https://www.gnu.org/licenses/gpl-3.0.txt

--2026-10-17 13:21:04--  https://www.gnu.org/licenses/gpl-3.0.txt
Resolving www.gnu.org (www.gnu.org)... 209.51.188.116
Connecting to www.gnu.org (www.gnu.org)|209.51.188.116|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 35149 (34K) [text/plain]
Saving to: 'gpl-3.0.txt'

gpl-3.0.txt                      100%[=============================================================>]  34.33K   157KB/s    in 0.2s

2026-10-17 13:21:06 (157 KB/s) - 'gpl-3.0.txt' saved [35149/35149]

rohitbhupat@Rohits-MacBook-Air sahil_devops % ls -ltr
total 80
-rw-r--r--@ 1 rohitbhupat  staff  35149 30 Sep  2017 gpl-3.0.txt
drwxr-xr-x  2 rohitbhupat  staff     64 16 Oct 10:33 Linux_Basics_Sahil_Shaikh
-rw-r--r--  1 rohitbhupat  staff    352 16 Oct 10:47 test_dir.zip
drwxr-xr-x  3 rohitbhupat  staff     96 16 Oct 10:48 test_dir
drwxr-xr-x  3 rohitbhupat  staff     96 16 Oct 10:50 unzipped_dir
rohitbhupat@Rohits-MacBook-Air sahil_devops %
```

6. Changing Permissions

1. Create a file named secure.txt and change its permissions to read-only for everyone using `chmod`.

```
rohitbhupat@Rohits-MacBook-Air sahil_devops % touch secure.txt
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops % ls -ltr
total 80
-rw-r--r--@ 1 rohitbhupat  staff  35149 30 Sep  2017 gpl-3.0.txt
drwxr-xr-x  2 rohitbhupat  staff     64 16 Oct 10:33 Linux_Basics_Sahil_Shaikh
-rw-r--r--  1 rohitbhupat  staff    352 16 Oct 10:47 test_dir.zip
drwxr-xr-x  3 rohitbhupat  staff     96 16 Oct 10:48 test_dir
drwxr-xr-x  3 rohitbhupat  staff     96 16 Oct 10:50 unzipped_dir
-rw-r--r--@ 1 rohitbhupat  staff      0 17 Oct 13:22 secure.txt
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops % ls -ltr | grep "secure.txt"
-rw-r--r--@ 1 rohitbhupat  staff      0 17 Oct 13:22 secure.txt
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops % chmod 400 secure.txt
rohitbhupat@Rohits-MacBook-Air sahil_devops %
rohitbhupat@Rohits-MacBook-Air sahil_devops % ls -ltr | grep "secure.txt"
-r--------@ 1 rohitbhupat  staff      0 17 Oct 13:22 secure.txt
rohitbhupat@Rohits-MacBook-Air sahil_devops %
```

7. Working with Environment Variables

1. Use `export` to set a new environment variable called MY_VAR with the value “Hello, Linux!”.

